

A-level

# ENVIRONMENTAL SCIENCE

## Paper 1 - Generated practice paper Set 05

### Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

### Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 01.1     | <i>Correct answer only.</i><br>• B - rivers are accessible over a large area with few natural contaminants (1)                                                                                                                                                                                                                                                                                                                                                                       | 1     |
| 01.2     | <i>1 mark for each point.</i><br>• Potable / public supply: no pathogens, low toxins, good appearance / taste / smell (public health) (1)<br>• Industrial boiler / cooling water: soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange; spray irrigation: low turbidity to avoid blocked pipes and low heavy metals (1)                                                                                                                        | 2     |
| 01.3     | <i>1 mark for each point.</i><br>• Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers) (1)<br>• Disadvantage: high cost / energy for pumping; ecological impacts on the donor river; transfer of pollutants or invasive species; political conflict (1)                                                                                                                                                    | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5.                                                                                                                                                                                                                                                                                                                                                      |       |
| 02.1     | <i>Any 4 from the points below - 1 mark each, maximum 4.</i><br>• Sludge from primary sedimentation is digested by anaerobic microbes in a warm tank for about four weeks (1)<br>• Kills most pathogens (1)<br>• Reduces odour and volume of the sludge (1)<br>• Produces biogas (mainly methane with some CO <sub>2</sub> ) used as fuel to heat the digester / generate electricity (1)<br>• Digested sludge has a better C:N ratio and can be used as agricultural fertiliser (1) | 4     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>02.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Heavy metals from urban wastes / road runoff may be present and absorbed by crops - reduced by adding lime to lower metal solubility <b>(1)</b></li> <li>• Remaining pathogens - destroyed by heat treatment of the sludge before use <b>(1)</b></li> <li>• Timing: spreading in the growing season covers crops; in winter vehicles compact wet soils - apply at suitable times / when soils are dry <b>(1)</b></li> <li>• Odour / public objection - digestion reduces odour <b>(1)</b></li> </ul>                                                                                                                                                                                                             | 4     |
| <b>02.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Foul waste is a small volume needing full treatment so smaller, cheaper treatment works can be built; surface water needs little treatment <b>(1)</b></li> <li>• The treatment works is not affected by weather-related fluctuations in flow (storms) so storm overflows of untreated sewage are avoided <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                   | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 270-273; AQA spec 3.4.3.2.9.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |
| <b>03.1</b> | <p><i>Correct answer only.</i></p> <ul style="list-style-type: none"> <li>• B - Low liposolubility and low persistence <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1     |
| <b>03.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Bioaccumulation: build-up of a pollutant in the tissues of an individual organism over time because uptake is faster than excretion / breakdown <b>(1)</b></li> <li>• Biomagnification: increase in concentration of the pollutant at successive trophic levels of a food chain <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                            | 2     |
| <b>03.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• They are intended to be toxic (to the target species) <b>(1)</b></li> <li>• They are deliberately released / spread into the environment (rather than being an accidental or waste release) <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |
| <b>04.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Instability / landslips on the hillside, especially after heavy rain - compaction, drainage, establishing vegetation whose roots bind the spoil, regrading slopes <b>(1)</b></li> <li>• Acid drainage from oxidation of pyrites, with toxic metals - collect and treat leachate, add lime to raise pH <b>(1)</b></li> <li>• Spontaneous combustion of residual coal / hydrocarbons - layers of fine-grained material to reduce air flow <b>(1)</b></li> <li>• Lack of nutrients / poor plant growth - add topsoil, sewage sludge, fertiliser <b>(1)</b></li> <li>• Unattractive topography - landscaping / reshaping <b>(1)</b></li> <li>• Dust / heavy metals - cover and keep at high pH <b>(1)</b></li> </ul> | 6     |
| <b>04.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• So that plant roots cannot reach the toxic layer <b>(1)</b></li> <li>• Otherwise toxins could be taken up and brought to the surface / dispersed via leaf litter and food chains <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2     |
| <b>04.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Steel slag is alkaline, producing unusual calcareous / high-pH soil conditions <b>(1)</b></li> <li>• Supports uncommon plants such as cowslips / orchids that cannot compete on fertile neutral soils <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 282-283; AQA spec 3.4.3.2.12.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |
| <b>05.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Inorganic mercury settled into anaerobic sediments where anaerobic microbes converted it to organic (methyl) mercury <b>(1)</b></li> <li>Methyl mercury is liposoluble so it is absorbed easily and stored in fat / passes through cell membranes <b>(1)</b></li> <li>It bioaccumulated in fish and shellfish and biomagnified along the food chain to reach much higher concentrations than in the water <b>(1)</b></li> <li>Local people ate large amounts of fish (critical group) so received toxic doses <b>(1)</b></li> </ul>                                                                                                                                                                 | 4     |
| <b>05.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Liposoluble so it crosses the blood-brain barrier into the brain, inhibiting enzymes / nerve function (neurotoxin) <b>(1)</b></li> <li>It crosses the placenta and harms the developing unborn baby <b>(1)</b></li> <li>Mercury is a teratogen, preventing normal gene expression / development <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                               | 3     |
| <b>05.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Elemental (liquid) mercury: not easily absorbed through skin or gut, though vapour is absorbed in the lungs <b>(1)</b></li> <li>Inorganic compounds (eg mercury oxide): moderately well absorbed in the gut <b>(1)</b></li> <li>Organic compounds (eg methyl mercury): easily absorbed through skin, gut and lungs; liposoluble so most toxic and bioaccumulates <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                              | 3     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 276-277; AQA spec 3.4.3.2.11.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |
| <b>06.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Trays of the same soil (texture, compaction, organic matter, depth) with different percentage vegetation cover (eg 0, 25, 50, 75, 100% grass), replicated <b>(1)</b></li> <li>Same slope / gradient; simulated rain from a standard height at a standard rate and drop size (eg watering can / sprinkler) for a standard time <b>(1)</b></li> <li>Collect the soil splashed / washed off (eg onto a sheet or into a trough at the base), dry and weigh it <b>(1)</b></li> <li>Repeat each treatment several times and calculate means (and standard deviation) <b>(1)</b></li> <li>Plot soil loss against cover; statistical test (Spearman's rank / t-test) for significance <b>(1)</b></li> </ul> | 5     |
| <b>06.2</b> | <p><i>Any 3 from the points below - 1 mark each, maximum 3.</i></p> <ul style="list-style-type: none"> <li>Rainfall intensity / raindrop size / height <b>(1)</b></li> <li>Soil texture, compaction, organic matter, depth <b>(1)</b></li> <li>Permeability of the material beneath <b>(1)</b></li> <li>Gradient <b>(1)</b></li> <li>Root binding <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                         | 3     |
| <b>06.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>All variables except the independent variable can be standardised in the laboratory; in the field uncontrollable variables (weather) must be measured instead <b>(1)</b></li> <li>Laboratory conditions may not represent natural conditions so results may not apply directly to real ecosystems <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                             | 2     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 419-420; AQA spec 3.7.1, 3.7.2.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |
| <b>07.1</b> | <i>Correct answer only.</i><br>• A waste is assumed to cause pollution until research shows it is unlikely to do so <b>(1)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1     |
| <b>07.2</b> | <i>1 mark for each point.</i><br>• If the cost of preventing damage is less than the (penalty for) damage there is a financial incentive to prevent pollution <b>(1)</b><br>• If polluters can escape their responsibilities / penalties are low, there is little incentive to comply <b>(1)</b>                                                                                                                                                                                                                                                                                                                                          | 2     |
| <b>07.3</b> | <i>Any 2 from the points below - 1 mark each, maximum 2.</i><br>• Marine discharges where currents dilute and disperse emissions <b>(1)</b><br>• Emissions downwind of urban areas <b>(1)</b><br>• Not discharging waste onto permeable rock above an aquifer <b>(1)</b>                                                                                                                                                                                                                                                                                                                                                                  | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-244; AQA spec 3.4.3.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |
| <b>08.1</b> | <i>1 mark for each point.</i><br>• Negative feedback: a change causes other changes that reduce the rate or size of the original change, re-establishing equilibrium <b>(1)</b><br>• Positive feedback: a change causes other changes that increase the original change <b>(1)</b>                                                                                                                                                                                                                                                                                                                                                        | 2     |
| <b>08.2</b> | <i>2 marks for each point (partial credit for a partly correct point).</i><br>• Increased low-level cloud: higher temperatures increase evaporation and condensation, producing more cloud; clouds have a high albedo so more sunlight is reflected and warming is reduced <b>(2)</b><br>• Increased photosynthesis: higher temperatures and CO <sub>2</sub> increase photosynthesis, removing CO <sub>2</sub> ; if stored in woody tissue atmospheric CO <sub>2</sub> rises less <b>(2)</b>                                                                                                                                              | 4     |
| <b>08.3</b> | <i>1 mark for each point.</i><br>• Permafrost: frozen waterlogged Arctic soils contain dead organic matter that decomposed anaerobically producing methane trapped in the frozen ground <b>(1)</b><br>• Warming thaws the permafrost, releasing methane (a powerful greenhouse gas) and allowing further decomposition, causing more warming <b>(1)</b><br>• Methane hydrate: methane from decomposed organic matter in deep sea sediments is held as a solid under high pressure and low temperature <b>(1)</b><br>• If deep water warms the hydrate melts, releasing methane into the atmosphere and causing further warming <b>(1)</b> | 4     |
| <b>08.4</b> | <i>Any 3 from the points below - 1 mark each, maximum 3.</i><br>• Reduced albedo as snow / ice melts <b>(1)</b><br>• Ocean acidification reducing coral survival and carbon sequestration <b>(1)</b><br>• Increased forest and peat fires releasing CO <sub>2</sub> <b>(1)</b><br>• Faster decomposition of soil organic matter releasing CO <sub>2</sub> <b>(1)</b><br>• Increased water vapour (a greenhouse gas) in warmer air <b>(1)</b><br>• Increased cirrus (high) cloud which traps infrared <b>(1)</b>                                                                                                                           | 3     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks |
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| <b>08.5</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• A point at which natural processes triggered by human-caused warming become self-sustaining / unstoppable <b>(1)</b></li> <li>• So that stopping the original human activity would no longer stop climate change (eg runaway permafrost melt, methane hydrate release, forest fires) <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-104, 371-372; AQA spec 3.2.1.2, 3.6.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |
| <b>09.1</b> | <p><i>2 marks for each point (partial credit for a partly correct point).</i></p> <ul style="list-style-type: none"> <li>• Ploughing: increases the oxygen supply to soil decomposers so aerobic decomposition of dead organic matter is faster; carbon in the soil DOM store falls and atmospheric CO<sub>2</sub> rises <b>(2)</b></li> <li>• Draining: waterlogged anaerobic conditions preserved dead organic matter (peat) and produced methane slowly; drainage makes the soil aerobic so stored carbon is rapidly decomposed to CO<sub>2</sub> (though methane release stops) <b>(2)</b></li> <li>• Marine pollution: kills phytoplankton so less dissolved CO<sub>2</sub> is absorbed by photosynthesis and less carbon sinks to sediments in dead plankton; less CO<sub>2</sub> then dissolves from the atmosphere to replace it, so atmospheric CO<sub>2</sub> rises <b>(2)</b></li> </ul> | 6     |
| <b>09.2</b> | <p><i>Any 3 from the points below - 1 mark each, maximum 3.</i></p> <ul style="list-style-type: none"> <li>• Rice paddy fields <b>(1)</b></li> <li>• Landfill sites <b>(1)</b></li> <li>• Anaerobic sediments in reservoirs <b>(1)</b></li> <li>• Livestock intestines (cattle) <b>(1)</b></li> <li>• Manure / slurry stores <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3     |
| <b>09.3</b> | <p><i>1 mark.</i></p> <ul style="list-style-type: none"> <li>• More CO<sub>2</sub> dissolves in the sea, forming carbonic acid which dissociates into hydrogen carbonate and hydrogen ions; more H<sup>+</sup> lowers the pH <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 150-151; AQA spec 3.2.4.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |
| <b>10.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Together they form a dispersed layer in the stratosphere (the ozone layer / ozonosphere) <b>(1)</b></li> <li>• They absorb ultraviolet light, which drives a dynamic equilibrium of reactions that form and destroy ozone <b>(1)</b></li> <li>• The UV energy is converted to chemical / heat energy so most UV that passed the upper atmosphere never reaches the surface <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                      | 3     |
| <b>10.2</b> | <p><i>Any 2 from the points below - 1 mark each, maximum 2.</i></p> <ul style="list-style-type: none"> <li>• Provides carbon, oxygen, hydrogen and nitrogen (as N<sub>2</sub>, O<sub>2</sub>, CO<sub>2</sub>, H<sub>2</sub>O) for biological molecules <b>(1)</b></li> <li>• Delays the escape of infrared energy, keeping the surface warm <b>(1)</b></li> <li>• Distributes heat and water vapour by winds / ocean currents <b>(1)</b></li> <li>• Provides atmospheric pressure so liquid water can exist <b>(1)</b></li> <li>• Deflects damaging radiation of the solar wind in the upper atmosphere <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                       | 2     |
| <b>10.3</b> | <p><i>1 mark.</i></p> <ul style="list-style-type: none"> <li>• Energy is absorbed at different altitudes (eg UV absorbed in the stratosphere, surface heating warms the troposphere) so temperature gradients change with altitude, producing layers <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks |
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| 10.4     | <p><b>Levels of response</b></p> <p><b>Level 3 (7-9 marks):</b> Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.</p> <p><b>Level 2 (4-6 marks):</b> Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.</p> <p><b>Level 1 (1-3 marks):</b> Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (credit other relevant, accurate points)</p> <ul style="list-style-type: none"> <li>• Troposphere: enhanced greenhouse effect - anthropogenic CO<sub>2</sub> (fossil fuels, deforestation), methane (rice, livestock, landfill), N<sub>2</sub>O, CFCs, tropospheric ozone; more infrared absorbed and re-emitted; aerosols and smoke increase albedo / scatter light; land-use change alters surface albedo and evapotranspiration</li> <li>• Stratosphere: CFCs / halons / HCFCs release chlorine and bromine which catalyse ozone destruction, shifting the O / O<sub>2</sub> / O<sub>3</sub> equilibrium; less UV absorbed in the stratosphere, more reaches the surface; stratospheric cooling</li> <li>• Consequences: changes to climate, ocean currents (thermohaline), hydrological cycle, sea level, species distributions; increased UV damage to organisms</li> <li>• Difficulties of prediction: linked processes where one change alters others; feedback (positive and negative) and tipping points; time lags (ocean heat capacity, long residence times of CFCs and CO<sub>2</sub>); natural variability (solar output, volcanoes, Milankovitch cycles); incomplete understanding of atmospheric processes; limited historical data so proxies needed; models with different assumptions</li> <li>• Evaluation: some effects well understood (greenhouse physics, ozone chemistry proven, Montreal Protocol success) but scale and timing of regional impacts uncertain; precautionary principle</li> <li>• Justified conclusion</li> </ul> | 9     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 86-89, 108-113; AQA spec 3.2.1.1, 3.2.1.3.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks |
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| 11.1     | <p><b>Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.</b></p> <p><b>Levels of response</b></p> <p><b>Level 5 (21-25 marks):</b> Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.</p> <p><b>Level 4 (16-20 marks):</b> Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.</p> <p><b>Level 3 (11-15 marks):</b> Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.</p> <p><b>Level 2 (6-10 marks):</b> Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.</p> <p><b>Level 1 (1-5 marks):</b> Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (students are not expected to cover all of these; credit other relevant material)</p> <ul style="list-style-type: none"> <li>• Properties: energy density, renewability, abundance, location, intermittency / predictability, storage, transport, conversions, technology level, environmental impacts</li> <li>• Fossil fuels: choosing lower-impact fuels (gas over coal), desulfurisation, CCS, efficient combustion, siting power stations near cooling water and coal ports; new extraction technologies (directional drilling, secondary / tertiary recovery) and their impacts (fracking, tar sands)</li> <li>• Nuclear: high energy density reduces transport; waste properties (half-life) determine storage; thorium and fusion prospects</li> <li>• Renewables: matching resource to location (wind, HEP, solar, wave, tidal, geothermal, biofuel siting); intermittency managed by storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management and diversity; low-impact designs (helical turbines, VAWT, tidal lagoons)</li> <li>• Energy conservation reduces demand: transport, buildings, industry (CHP, heat exchangers)</li> <li>• Impacts before and during use; life-cycle / embodied energy</li> <li>• Evaluation: no resource is impact-free; knowledge allows the best mix and mitigation; judgement</li> </ul> <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.</p> | 25    |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks |
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| 11.2     | <p><b>Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.</b></p> <p><b>Levels of response</b></p> <p><b>Level 5 (21-25 marks):</b> Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.</p> <p><b>Level 4 (16-20 marks):</b> Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.</p> <p><b>Level 3 (11-15 marks):</b> Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.</p> <p><b>Level 2 (6-10 marks):</b> Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.</p> <p><b>Level 1 (1-5 marks):</b> Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (students are not expected to cover all of these; credit other relevant material)</p> <ul style="list-style-type: none"> <li>• Composition and structure: N<sub>2</sub>, O<sub>2</sub>, Ar, CO<sub>2</sub>, ozone; troposphere and stratosphere; pressure and liquid water; gases for biological molecules and industry</li> <li>• Energy processes: UV absorbed by O / O<sub>2</sub> / O<sub>3</sub> dynamic equilibrium; visible light absorbed at the surface; infrared absorbed by greenhouse gases (33 °C warming); heat distribution by winds and ocean currents; transport of water vapour; energy budget and albedo</li> <li>• Global climate change: anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species; difficulties of monitoring and prediction; control strategies and agreements</li> <li>• Ozone depletion: CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time</li> <li>• Other threats: aerosols, smoke, land-use change; interconnection of processes</li> <li>• Evaluation: which functions are most at risk; success of ozone controls vs slow progress on greenhouse gases; justified conclusion</li> </ul> <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 86-113; AQA spec 3.2.1.</p> | 25    |